

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Defined core infrastructure using **Terraform IaC** – multi-AZ **ECS Fargate** with explicit lifecycle management (create, update, scale, destroy); reproducible, version-controlled environments; **99.9% uptime SLO**, **44% cost reduction** (\$40–60/month); auto-rollback on health check failure; **vLLM inference engine with PagedAttention** for zero-OOM ML serving at production traffic
- Built **observability and distributed tracing stack** – **CloudWatch** for infra health, **Prometheus + Grafana** for application metrics (eval success rate, OOM rate, gRPC latency), **gRPC inter-service tracing** for production debugging; on-call rotation with runbooks and post-mortems; zero undetected pipeline failures
- Designed gRPC inter-service layer for 3-person engineering team – debugged production deadlock by tracing shared resource acquisition conflicts and redesigning call sequencing; **exactly-once delivery** across distributed services; circuit breaker pattern with auto-rollback; **sub-50ms p99** at production traffic
- Led backend development – **Python (FastAPI/Django)** REST APIs, **PostgreSQL + Redis** data layer, **LLM API integration** (Claude, GPT); LLM-as-a-judge evaluation pipeline for ML correctness metrics; **AWS/GCP multi-cloud** (ECS Fargate, EKS, EC2, S3, RDS)

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with **Protocol Buffers** for low-latency inter-process communication – selected Protocol Buffers for schema evolution and cross-language compatibility; shipped to Chrome stable at **sub-50ms p99**, **10K+ req/sec**, **3B+ active users**
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern – **68% feature delivery acceleration** across **25K+ lines** of Chromium at **95% test coverage**; shipped to **3B+ active users**
- Delivered design documents and code reviews at **95% test coverage**; changes adopted into production branch reviewed by senior Chrome engineers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed **staging environment** with automated integration testing – **90% deployment time reduction** via CI/CD (GitHub Actions); JWT stateless BaaS on AWS supporting **500+ concurrent users** with horizontal auto-scaling
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with **PostgreSQL composite indexing** achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC (AWS/Azure/GCP)

github.com/pulumi

- Submitted **Go CLI features** enabling reproducible **multi-cloud (AWS/Azure/GCP) infrastructure provisioning** – developer tooling for infrastructure-as-code under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), GCP, **Terraform**, **Kubernetes**, Docker, CI/CD, CloudWatch, Prometheus, Grafana

Distributed Systems: gRPC, Protocol Buffers, Exactly-once semantics, Circuit breakers, Fault tolerance, Multi-AZ, Event-driven architecture

Languages: Python, Go, TypeScript, C++, Java, JavaScript, SQL, Bash, Rust

Frameworks & Databases: FastAPI, Django, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM APIs (Claude, GPT), LangChain, LLM-as-a-judge evaluation, PyTorch

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor